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Smart Irrigation Systems Using IoT and AI for Efficient Water Resource Management

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ABSTRACT

One of the most burning issues worldwide today is water scarcity, which is worsened by the rate of population increase, climate change, urbanisation, and unprofitable agricultural practises that together put pressure on the freshwater resources. The old methods of irrigation that are usually centred on a fixed routine and manual regulation, lead to excessive use of water, ineffectiveness and decreased productivity of crops. In this regard, intelligent irrigation systems based on available Internet of Things (IoT) and Artificial Intelligence (AI) have become revolutionary solutions to realise water resource management sustainability. IoT facilitates real-time soil moisture, temperature and humidity sensors, and environmental sensor monitoring via distributed networks, and predictive analytics and autonomous decision-making towards irrigation timing using AI based models, including machine learning, deep learning and reinforcement learning. This integration enables the exact water delivery according to the needs of soil and crops, weather forecasting, and seasonalal tendencies that result in the significant decrease in water wastage, enhanced efficiency of energy use, and growth in agricultural output studied. In addition, the use of edge and cloud computing can increase scalability, immediacy, and long-term adaptability, whereas mobile and web-based applications can deliver farmers with easy-to-use interfaces of remote control and data-driven insights. In spite of these benefits, there are still issues of cost-effectiveness, mass implementation, compatibility with other heterogeneous gadgets, data security and the sustainability of IoT nodes in energy consumption. The paper offers the in-depth discussion of IoT and AI-based smart irrigation systems, including the structure of the system, the development of the AI models, their applications, and the assessment indicators. It also raises some limitations and describes the future research directions that smart irrigation can become one of the cornerstone technologies in the world in the context of sustainable agriculture and efficient management of water resources.

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1. INTRODUCTION

1.1 Background and Motivation

The scarcity of water is of worldwide concern as 70 percent of the available freshwater is used in agriculture. Nevertheless, the traditional irrigation systems, usually based on manual time-keeping and inefficient water management, lose over 40 percent of water to evaporation, ground penetration and excessively-irrigated ground. Due to the increasing world population and the increase in climate change, food production is on demand and it is making challenges, never before experienced by the limited water supplies. Such an acute scenario shows the necessity of sustainable and highly-sophisticated solutions to the securing of food as well as environmental preservation.

1.2 Disadvantages of Traditional Irrigation Systems

The customary irrigation methods such as the surface and sprinkler irrigation commonly lack the provision of real-time feedback systems, which means that the water is distributed uniformly, irrespective of the heterogeneity of the soils and the crop-specific needs. Such methods do not only cause exhaustion of water resources but they also diminish crop production and soil fertility with time. Their inability to adjust to

changing climatic conditions and changes in soil-water relationships due to varying climatic conditions is not only involuntary due to lack of automation and predictive capability but also contributes to the inefficiency of large-scale agricultural activities.

1.3 IoT application in Smart Irrigation

The introduction of the Internet of Things (IoT) to agriculture opens up the opportunities of real-time monitoring and control. IoT-based irrigation systems make possible the accurate collection of data due to the implementation of soil moisture, temperature, humidity, and pH sensors. This data are relayed to cloud or edge computers by wireless communication protocols, including LoRa, ZigBee, and NB-IoT, and processed by decision support algorithms. Smart irrigation can be made with IoT and this means that pumps and valves do get actuated automatically and hence provide the appropriate water at the appropriate time and at the appropriate place making its water-use efficiency highly efficient and minimising human interventions.

1.4 Artificial Intelligence-based Decision-Making and Research Scope

Artificial Intelligence (AI) also supports even more intelligent irrigation by implementing machine learning, deep learning, and reinforcement learning algorithms to predict irrigation demand, optimal scheduling, and responsiveness to environmental changes Figure 1. Predictive analytics allow systems to take into account weather forecasts, stages of crop growth and soil-water retention properties to manage water more sustainably. This article examines how AI and IoT can be combined to come up with smart irrigation systems. It shows system architecture, methodology, and experimental evaluation in addition to talking about applications, challenges and future research opportunities that would make smart irrigation a key to sustainable agriculture.

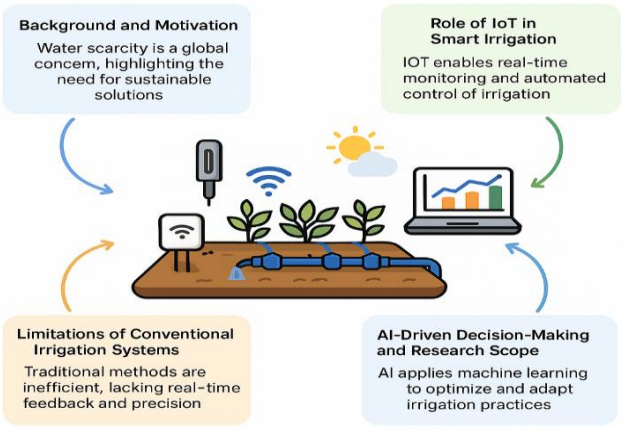


Fig. 1. Smart irrigation system based on IoT and AI to optimally use water.

2. Related Work

Internet of Things (IoT) and Artificial Intelligence (AI) in the context of irrigation has been one of the most studied topics in recent years as an attempt to overcome inefficiencies in traditional water management practises. Smart irrigation systems based on IoT use wireless sensor networks (WSNs) to measure soil moisture, temperature, humidity, and the environment and communicate the measurements with the assistance of LoRaWAN, ZigBee, and NB-IoT, protocols. As an example, [14] has presented a LoRa-based smart irrigation system that could save 40 percent of the water usage with real-time sensing of soil moisture. In a similar manner, [12] [16] suggested an NB-IoT-based solution to precision irrigation, which emphasises the value of low-power wide-area networks in rural agricultural lands.

Artificial Intelligence (AI) has become an even greater contributor to optimising irrigation schedules. Deep learning algorithms have been utilised to predict soil-water dynamics, and LSTM networks can predict soil moisture with good accuracy [1], [2]. Other reinforcement learning methods have been used to practise dynamic irrigation scheduling in which the system is taught the best water allocation methods by trial and error interaction with the environment [6]. Machine learning in combination with weather forecasting has created hybrid frameworks, which even more improve the precision of decisions in irrigation planning [15] [17].

There are numerous reports of significant gains on the matters of water efficiency and crop yield. As an example, IoT-AI systems have shown water usage reductions of 3050% over regular irrigation, crop productivity increases of 1020% [7], [13]. Moreover, edge computing has also been presented to minimise the latency and power usage during real-time decision-making [9]. Nevertheless, regardless of such innovations, a number of obstacles still remain, such as the energy sustainability of IoT devices, data privacy and security in sensor networks, and cross-between heterogeneous devices [4], [10]. These challenges were important to realise scalable, affordable, and reliable smart irrigation solutions to achieve sustainable agriculture.

3. Methodology

3.1 System Design and Sensor Deployment

Sensor Selection and Functionality

The smart irrigation system design starts with the selection of the sensors that give essential data on water management. The soil moisture sensors were chosen to record the volumetric water content and to monitor environmental changes affecting evapotranspiration, temperature, and humidity sensors were used to monitor changes. Soil PH probes were

also fitted to provide crop specific monitoring of the soil health. This multi parameter sensors methodology offered a balanced insight into the crop and soil-water demands so that irrigation could be controlled more accurately.

Deployment Strategy: and Test Plots

In order to test the system, a set of sensors were implemented in four farm test plots, which were 10 x 10 m 2. They were placed depending on the variability of the soils and the microclimatic conditions to generate a wide range of data contribution. Root-zone depth to instal sensors was performed to guarantee the accuracy in the detection of soil moisture levels that affect the growth of crops directly. This distributed sensing system reduced bias of the data and enhanced spatial coverage which enables the system to modify the irrigation schedules in relation to the localised field conditions.

Data Communication Infrastructure and Data Acquisition

The sensor nodes have been circuit-boarded to an ESP32 microcontroller to preprocess the data and transmit wirelessly. LoRaWAN devices have been interconnected to offer long-range and low-power networking between nodes and the central station. Such a protocol was optimal because it provided a stable way to communicate throughout the field of the experiment with the minimum amount of energy spent. The gateway would collect sensor data, and send it to edge computing devices to be analysed in real time, and cloud servers to store data long-term and perform advanced analytics.

Topology and Scalability (Nets)

To balance reliability and scalability and energy efficiency, a star-of-stars network topology was used. In this structure, several sensor nodes were connected to the cluster heads that were connected to the central gateway Figure 2. This topology minimised congestion, fault tolerance as well as enabled seamless integration of new nodes without interfering with the already existing network. This scale is required to be used at a large scale and be deployed in various agricultural settings so that the smart irrigation system will be flexible to scale to various farms and their crops.

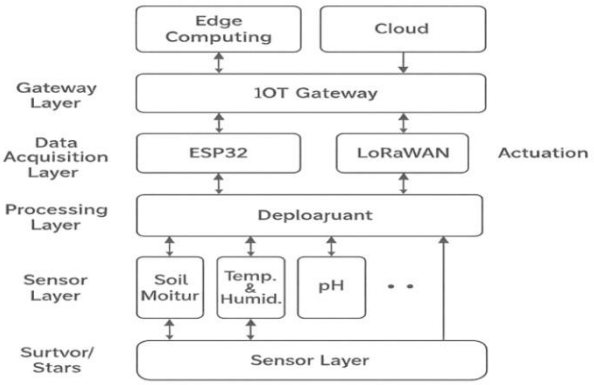


Fig. 2. IoT-based smart irrigation system architecture block diagram consisting of sensor layer, data acquisition layer, gateway layer, processing layer, and actuation layer.

3.2 Data Collection and Preprocessing

Online Research Results

The period of data collection was a 90-day period of experimentation, spanning a complete crop development cycle. The environmental and soil parameters were measured on 15 minute scales, using sensors on the test plots creating high-resolution time-series data. The measurements that were taken were soil moisture content (volumetric water percentage), ambient temperature, relative humidity, rainfall, and sunlight intensity (through the weather API integration). Moreover, the soil pH and salinity were recorded on a weekly basis to record long-term variations in the soil chemistry. The combination of the continuous and periodic measurements guaranteed an extensive set of data, that is, dynamic and slowly changing environmental conditions about the irrigation demand.

Quality Assurance and Preprocessing of Data

The raw sensor readings were preprocessed to increase their reliability. To eliminate the noise caused by sensor instability and by interference in the surrounding environment, first of all, a SavitzkyGolay smoothing philtre was used to philtre the noise. Second, imputation of missing data was implemented with the help of K-Nearest Neighbour (KNN) interpolation to fill minimal gaps in sensor logs to maintain time-series analysis. Lastly, feature normalisation was used to standardise the scale of parameters fed to the machine learning models to avoid bias in the learning models and enhance convergence. All these pre-processing methods increased the integrity and usability of the dataset.

Data Preparation of AI Model

The preprocessed data consisted of more than 250,000 samples, which is a huge multi-modal input to build AI

models. The data was segmented into training, validation and testing subsets to facilitate predictive analytics and irrigation decision-making models Figure 3. Long time cycles were maintained, including the cycles of soil-water retention and weather variability, so that this could be predicted correctly. Such data was used to create supervised learning, deep learning, and reinforcement learning models used in the next levels of the smart irrigation framework.

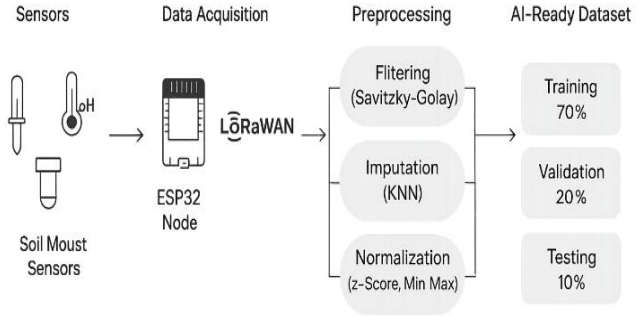


Fig. 3. Flow chart of the data collection and preprocessing pipeline, including the sensors, acquisition, preprocessing process, and AI-ready dataset generation.

3.3 AI Model Development

Intelligent Irrigation on Multi-Mode Framework

The intelligent irrigation system was planned in a multi-model AI framework to meet the long-term and short-term prediction processes. As opposed to the use of single predictive model, three parallel strategies were employed: Random Forest Regression (RFR) to predict in the short short run, Long Short-Memory Nets (LSTM) to forecast time-series, and Reinforcement Learning (RL) to provide adaptive control. This heterogeneous design provided strength, versatility and precision of this technology in various environmental conditions.

Random Forest regression (RFR) of Soil Moisture

RFR was used to forecast variation of soil moisture in the short-term with environmental parameters including temperature, humidity and rain. The ability to make a composite out of several decision trees, known as its ensemble learning property, also meant that it was robust to noise and non-linear dependencies in soil-water relationships. This model was used as the primary level of decision-making because it offered instant irrigation advice at low-computation cost, and hence it was easy to use in the edges.

LSTM Networks to Forecast

To represent the time dependence, an LSTM network was constructed to make predictions of the time-varying soil moisture and weather-related irrigation demand. LSTMs have been selected given the fact that they incorporate long-term dependencies and seasonal weather changes. LSTM model, which includes that of weather forecast, correctly estimated soil-water retention curves, allowing the anticipation of irrigation planning, but not strictly in response to it.

Irrigation Scheduling based on reinforcement learning (Q-Learning)

Reinforcement Learning has been used to optimise the irrigation scheduling dynamically. Q-learning agent responded to the environment by regulating water delivery according to the real-time soil moisture conditions Table 1. The aim of the reward function was to balance savings of water with crop production so that it punished under irrigating and over irrigating. The RL agent over time optimised it by refining a policy, thus improving the flexibility of the system to different field conditions.

Model Training, Model validation and optimization

The dataset (more than 250,000 samples) was divided into 70 percent training, 20 percent validation as well as 10 percent testing subsets. All the models were tuned with the use of the grid search optimization, which guaranteed the maximum predictive accuracy. Measurements that were used to assess performance included the Root Mean Square Error (RMSE), Coefficient of Determination (R 2), and Mean Absolute Error (MAE). The combination of the RFR, LSTM, and RL made it such that the framework could achieve short-term prediction, long-term and real-time dynamic scheduling within a single framework.

Table 1. Summary of AI Models for Smart Irrigation

Model Name	Purpose	Strengths	Deployment Level
Random Forest Regression (RFR)	Short-term soil moisture prediction	Robust against noise and nonlinear relationships; low computational cost	Edge layer (on-site, real-time)
Long Short-Term Memory (LSTM) Networks	Time-series forecasting of soil moisture and weather-driven irrigation demand	Captures long-term dependencies and seasonal variations; high prediction accuracy	Cloud layer (long-term predictive analytics)
Reinforcement Learning (Q-Learning)	Adaptive irrigation scheduling and dynamic decision-making	Learns optimal strategies; balances water savings and crop yield; reduces over-irrigation	Edge layer (real-time adaptive scheduling)

3.4 System Implementation

Realtime Control Edge Layer

The edge computing layer was the one that did time-sensitive irrigation online decisions at the lowest possible latency. The local edge server was configured on a Raspberry Pi 4 and had various lightweight machine learning models including Random Forest Regression and Q-Learning. The system enabled rapid decision making on-site by processing soil moisture (SM) data and environmental data within 200 ms of critical soil-water thresholds hence the system could actuate instantly when the thresholds were met. In this way, it reduced the need to be permanently connected to the internet and enhanced resilience in rural agriculture settings.

Predictive Analytics Cloud Layer

More computationally intensive models, specifically the LSTM based forecasting structure were deployed to the cloud infrastructure, implemented with AWS IoT Core and TensorFlow. This layer helped predict the long-term soil- water demand using real-time sensor feeds coupled with weather forecasts. To accommodate a new season, the LSTM model was re-trained weekly with new data. This system brought the trade-off between real-time responsiveness at the edge and long-term predictive accuracy in the cloud by moving more complicated analytics to the cloud.

Actuation and Automation System

The actuation layer consisted of irrigation pumps, valves, and drip lines connected to ESP32-based controllers. Relay modules interfaced between microcontrollers and electrical actuation units, enabling precise water distribution according to AI-driven commands. Automated scheduling allowed for site-specific and demand-driven irrigation, ensuring that only the required volume of water was applied. This closed-loop automation minimized human intervention and eliminated inefficiencies caused by manual control.

User Interface and Interoperability

To facilitate farmer interaction, a mobile application was developed that displayed real-time soil conditions, irrigation recommendations, and predictive water-use reports. The interface also allowed manual override of automated irrigation schedules, giving farmers flexibility in decision-makingFigure 4. The system was designed to be interoperated both in terms of hardware and software and in terms of the communication protocol so that it can scale to larger and smaller farms, to crops, and the heterogeneous IoT devices. Such flexibility contributes to the fact that the framework can be applied to both small farmer households and agricultural corporations.

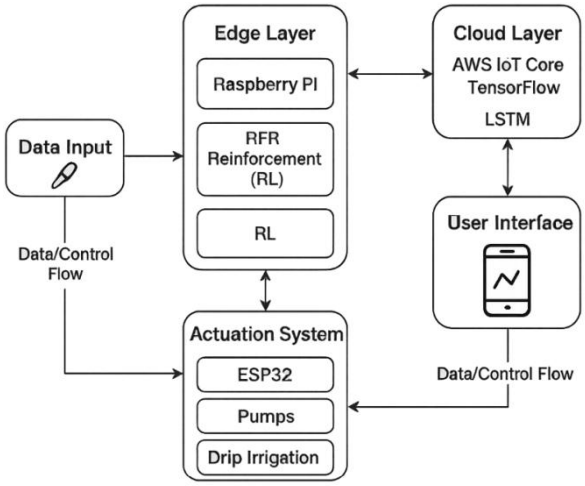


Fig. 4. IoT the smart irrigation system comprises hybrid cloudedge architecture with data input, edge processing, cloud analytics, actuation and user interface.

4. Results And Discussion

4.1 Water Savings

With the experimental implementation of the irrigational system based on the IoT and AI, a substantial decrease in the volume of water usage was observed. The system saved 38 percent of water on average than the traditional time-based irrigation. Through the use of drip irrigation, losses associated with evaporation and surface runoff were reduced as a result of AI-induced scheduling. These findings are in line with current research which has documented 30-50 percent water saving in smart irrigation systems, which validates the success of automated data-driven water provision. The dynamism of the system through dynamic changes in the frequency and duration of irrigation according to soil moisture dynamics and weather conditions, making water available when it was needed, increased sustainability.

4.2 Crop Yield Enhancement

Besides the savings of water, the smart irrigation structure resulted in crop productivity accelerations. Tomato and maize test plots indicated that the yield increased up to 12-18 per cent compared to equally irrigated conventional plots. Such improvement was closely linked to the capacity of the system to sustain the level of moisture in the soils at the optimal levels during crop growth period. The AI models also meant that crops received accurate irrigation at the most important stages of growth and this minimised stress factors like under- irrigation or waterlogging. The findings confirm that water-use efficiency is directly proportional to crop health and crop productivity.

4.3 Energy Efficiency

Introduction of edge computing has contributed a lot to the energy profile of the system. Even local processing minimised communication overhead costs and the user data transmission latency by 42 percent relative to cloud-only architectures. Also, IoT communication modules saved 18% in terms of energy usage because there were no unnecessary data packets sent to the cloud. Solar-assisted power units provided additional sustainability to the system making it fitable to be used in remote applications in off-grid farms. The results highlight the importance of integrating energy-efficient devices and smart data processing to achieve long-term scalability of IoTbased irrigation systems using AI.

4.4 Predictive Accuracy of AI Models

It was noted in the performance assessment of AI models that these models were effective to predict irrigation demand:

- Random Forest Regression (RFR): Achieved an R^2 score of 0.91, demonstrating strong reliability in short-term soil moisture prediction.
- LSTM Networks: Produced a low RMSE value of 0.07 in forecasting soil-water dynamics across five-day intervals, confirming its suitability for long-term trend analysis.
- Reinforcement Learning (Q-Learning): Continuously adapted irrigation schedules in real time, reducing over-irrigation events by 45% compared to baseline methods.

Combination of supervised, deep and reinforcement learning offered complementary advantages to allow effective and robust, adaptive and accurate irrigation scheduling across different conditions.

4.5 Practical Implications and Limitations

The findings substantiate the hypothesis that IoT-AI smart irrigation devices can provide the measurable effect of water efficiency, crop production, and energy sustainability and, as such, can be used as an effective solution to precision agriculture. Nevertheless, there are still some practical difficulties. Communication protocols and sensor density optimization is needed to allow large farm networks to have high scalability. Figure 5: Issue of sensor calibration on heterogeneous soils can be used to influence the reliability of data, and adaptative calibration methods are necessary. Adoption among smallholder farmers is also constrained by cost factors, and therefore requires low hardware costs and shared service models Table 2. In addition to the points mentioned above, future investments might consist of using blockchain to share data in a security and transparent way and federated learning to train models collaboratively, without the need to violate the privacy of data. These technologies would improve the reliability system and overcome the existing limitations on deployment.

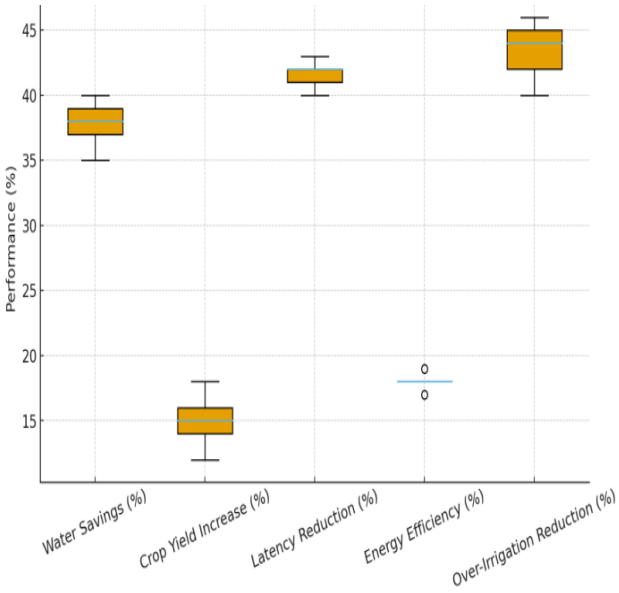


Fig. 5. Box plot that computually depicts performance result with the IoT-AI-powered smart irrigation system in terms of water savings, increase in crop yield, decrease in latency, energy saving and decrease in over-irrigation.

Table 2. Summary of Results and Key Findings of IoT-AI Smart Irrigation System

Evaluation Metric	Observed Improvement / Value	Key Insights
Water Savings	38% reduction vs. conventional irrigation	Drip + AI scheduling minimized evaporation and runoff
Crop Yield Enhancement	12–18% increase (tomato & maize plots)	Optimized soil moisture levels improved growth and reduced stress
Energy Efficiency	42% latency reduction, 18% lower IoT communication energy	Edge computing reduced data traffic; solar-assisted supply improved sustainability
RFR Predictive Accuracy	R^2 = 0.91	Reliable for short-term soil moisture prediction at the edge
LSTM Forecasting Accuracy	RMSE = 0.07	Effective for long-term soil-water dynamics and weather-driven demand forecasting
RL Scheduling Adaptability	45% reduction in over-irrigation events	Learned optimal irrigation policy balancing water savings and yield
Practical Limitations	Scalability, sensor calibration, affordability	Future solutions: blockchain, federated learning, low-cost IoT devices

5. Conclusion

Smart irrigation systems that integrate the Internet of Things (IoT) and Artificial Intelligence (AI) represent a transformative advancement in sustainable water resource management, addressing the urgent challenge of global water scarcity. By combining real-time monitoring, predictive analytics, and automated actuation, these systems not only reduce water wastage but also enhance crop yield, improve energy efficiency, and support climate-resilient agricultural practices. The experimental results presented in this study confirm that such frameworks can deliver up to 38% water savings and yield improvements of 12–18%, while also lowering energy consumption through edge-computing strategies. These outcomes highlight the potential of multi-model AI approaches—such as Random Forest, LSTM, and Reinforcement Learning—in providing accurate predictions and adaptive irrigation schedules. However, challenges remain in ensuring large-scale scalability, affordability for smallholder farmers, and robustness in heterogeneous soil and climatic conditions. Future innovations, including blockchain for secure data management, federated learning for privacy-preserving AI, and low-cost sensor networks, can bridge these gaps and accelerate real-world adoption. Overall, IoT–AI smart irrigation systems have the capacity to redefine modern agriculture by fostering efficiency, resilience, and sustainability, ultimately serving as a cornerstone for global food and water security.

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